



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Human–Computer Interaction — CA-3

Raktdaan: A Blood Donation and Blood Bank Mobile App — UI/UX Design in Figma with an HTML & CSS Prototype

Group Number: Group – 06

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PROBLEM STATEMENT

Finding blood still runs on phone calls and forwards

Patients' families

An urgent need turns into phone calls and forwarded posts. Nobody can tell who has seen the request or who is on the way.

Willing donors

They do not know which centre needs their blood group, whether they are eligible today, or when a slot is free.

Blood centres

Stock levels are hidden from donors, and donors rarely hear what happened to their blood — so fewer come back.

Our problem statement: design a mobile app that lets a willing donor find a centre, check eligibility and book a slot in a few taps, and lets a patient's family send an urgent request to nearby compatible donors — then follow every step live.

Scope of this CA-3

Figma design system

15 high-fidelity screens

Clickable prototype

HTML & CSS build

USERS

Who we designed for: three proto-personas

Aarav Deshpande · Donor

Primary user · B+ · Kothrud, Pune · 6 donations since 2021

- **Wants** to know when he can donate again and book without paperwork.
- **Wants** to see where his blood went.
- **Frustrated by** forms at every centre and no news after donating.

Meera's family · Requester

Primary user · needs 2 units of B+ at Sahyadri Hospital within 6 hours

- **Wants** to reach compatible donors fast, without knowing transfusion rules.
- **Wants** to know who is coming and when.
- **Frustrated by** panic, guesswork and no status updates.

Blood-centre staff

Secondary user · blood bank desk at a hospital

- **Wants** donors who arrive eligible and on time.
- **Wants** quick check-in and verified donor details.
- **Frustrated by** deferrals at the desk and repeated paperwork.

Proto-personas are assumption-based (Cooper, 1999). They guided our design decisions and should be validated with real donors and families in usability testing.

APPROACH

Design process and design goals

1 · UNDERSTAND

Problem, users, Indian donor rules (NBTC 2017)



2 · DEFINE

Proto-personas, goals, five user flows



3 · DESIGN

Figma variables, text styles, components, 15 screens



4 · PROTOTYPE

Figma prototype links, then HTML & CSS



5 · EVALUATE

Nielsen's heuristics, WCAG contrast and targets

Fast under stress

An urgent request fits on one screen; big primary buttons sit at the bottom, in thumb reach.

Trustworthy

Verified donors, government-licensed centres and stock that says when it was last updated.

Private by default

A hospital sees a donor's number only after the donor accepts the request.

Clear

One question per screen, plain words, and "Why do we ask this?" next to medical questions.

Motivating

A "ready to donate" countdown, an impact tracker and badges bring donors back.

Accessible

Large touch targets, colour never used alone, visible focus states.

15 screens in 5 flows, one bottom navigation

01

Onboarding

- 1.1 Welcome
- 1.2 Sign in
- 1.3 Verify OTP
- 1.4 Blood group

02

Home & discover

- 2.1 Home
- 2.2 Find centres (map)
- 2.3 Centre details
- 2.4 Eligibility check

03

Book a donation

- 3.1 Choose a slot
- 3.2 Booking confirmed

04

Emergency request

- 4.1 Request blood
- 4.2 Matching donors
- 4.3 Request tracking

05

Donor card & impact

- 5.1 Donor card
- 5.2 Your impact

Donor: Home → Find → Centre details → Choose a slot → Booked

Urgent donor: Home → “I can donate” → Eligibility check → Slot

Requester: Request → Matching donors → Request tracking

Home

Find

Request

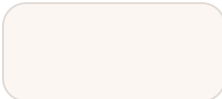
Donor card

Profile

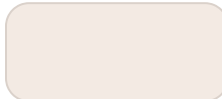
Request is a raised button in the centre of the bottom bar, so the emergency path is one tap from any main screen.

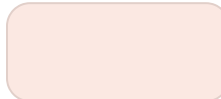
Foundations: colour, type and shape

BACKGROUNDS


bg-canvas
#FBF6F1


bg-surface
#FFFFFF


bg-sunken
#F3EAE3


bg-blush
#FBE8E2


bg-wine
#4A1D24

TEXT


text-primary
#2A1E1C


text-secondary
#6B5A55

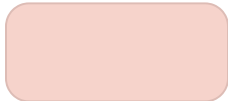

text-tertiary
#9A8983


text-brand
#A82E28

BRAND


brand
#C2362F

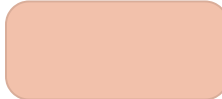

brand-pressed
#9E2A23


brand-soft
#F6D3CB

ACCENTS (STATUS)


sage
#3E7B5B


turmeric
#B7791A


peach
#F2C1AB


plum
#7A3346

FRAUNCES · DISPLAY & HEADINGS

**Be the reason someone gets a
tomorrow.**

Display 34/40 · Headline 26/32 · Title L 20/26 · Numeral 40/44

FIGTREE · BODY & INTERFACE

Find blood centres near you and book a slot in seconds.

Title 17/24 · Body L 16/24 · Body 14/20 · Label 15/20 · Caption 12/16 · Overline 11/14

CORNER RADIUS & ELEVATION


10


16


24

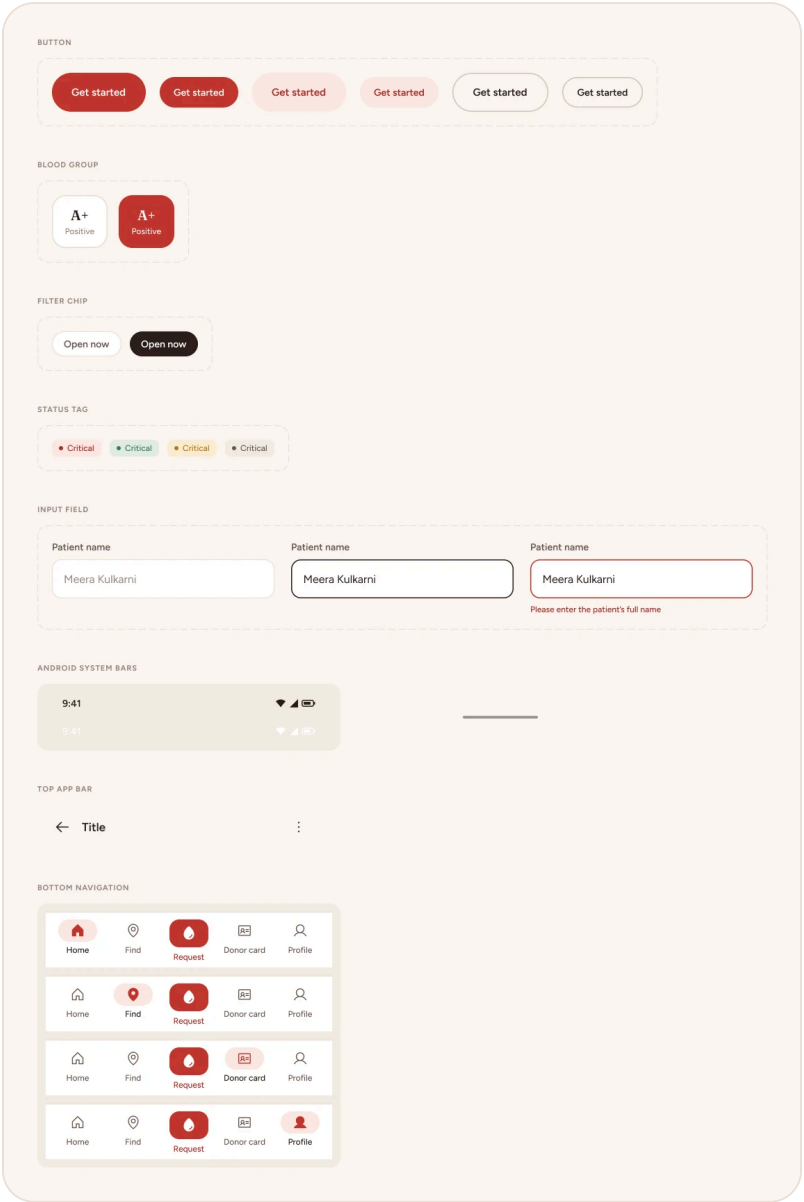

pill


soft


raised

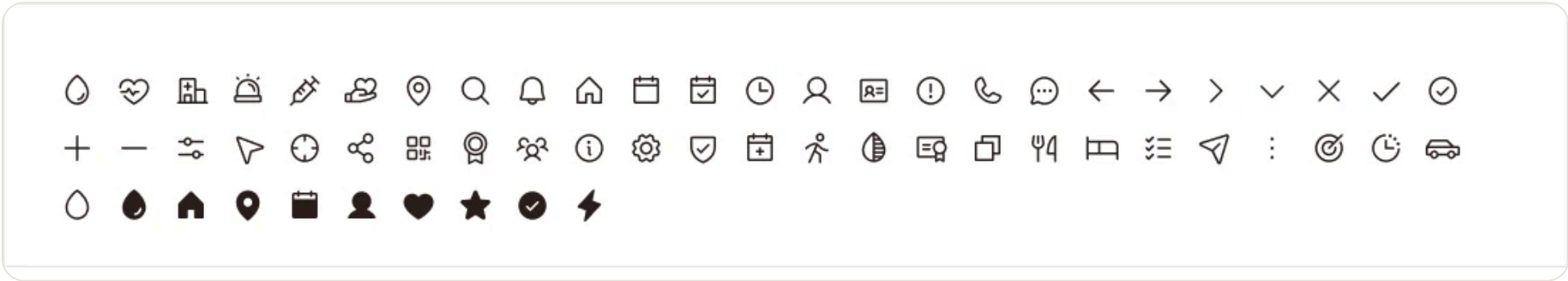
Warm neutrals keep the app calm; red is kept for actions and urgency. The same token names are used as Figma variables and as CSS custom properties.

Reusable components and one icon family



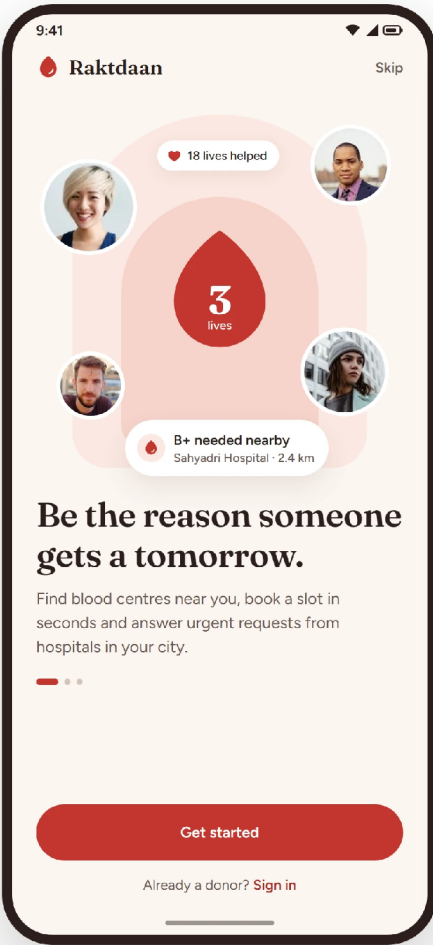
- **Button** — primary, secondary and outline, in 56 px and 44 px heights.
- **Blood-group chip** — default and selected; works as a radio button.
- **Filter chip** — off and on; works as a checkbox.
- **Status tag** — critical, ready, soon, neutral: a dot, a colour *and* a word.
- **Input field** — default, focused and error with helper text.
- **Android system bars, top app bar, bottom navigation** (four active states).

ICONS · PHOSPHOR, REGULAR + FILL

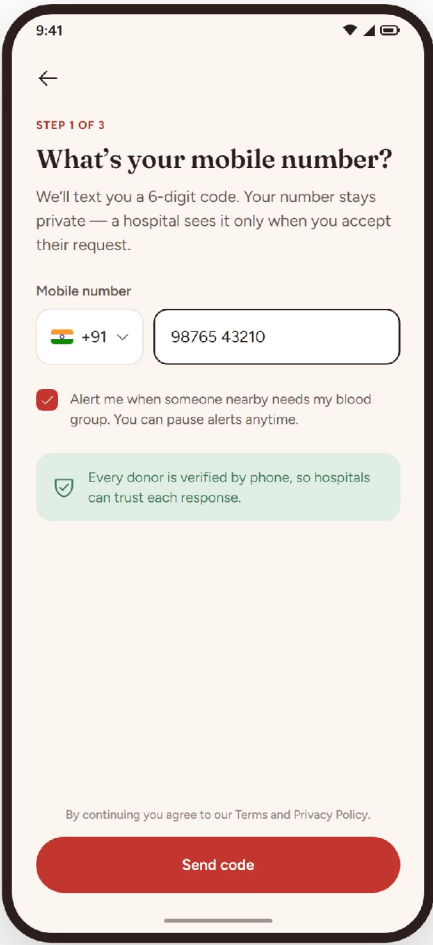


One stroke style everywhere; the filled version marks the active tab and selected states.

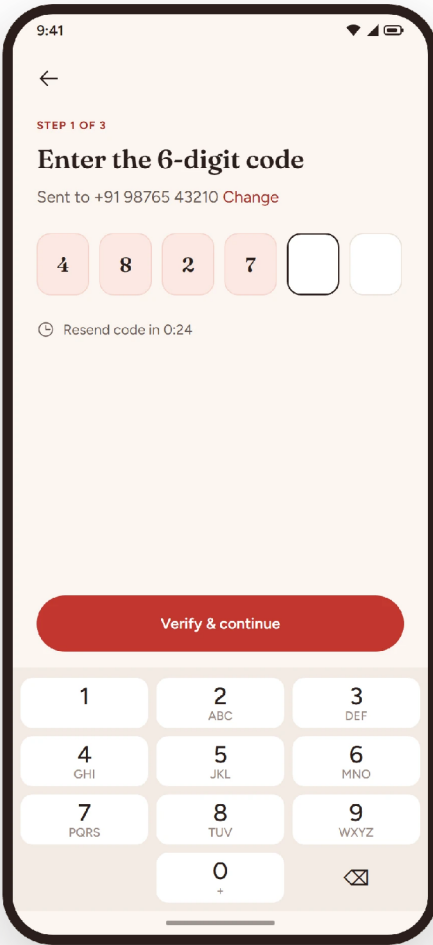
Sign up in three short steps



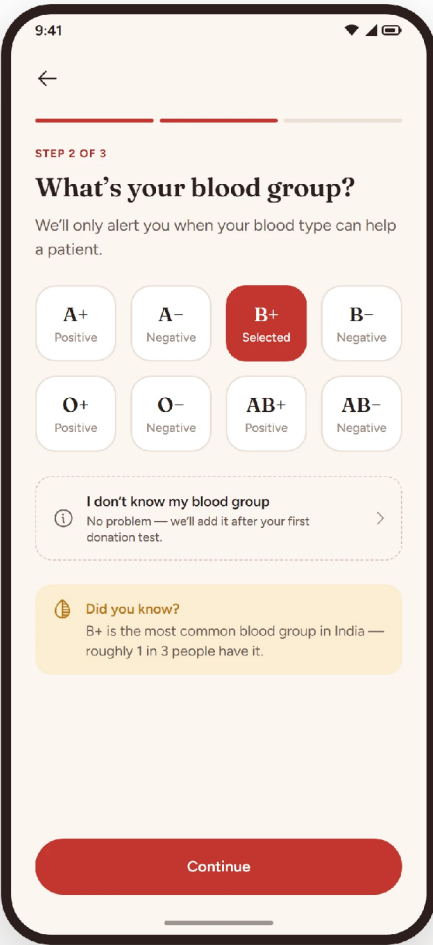
1.1 Welcome



1.2 Sign in



1.3 Verify OTP



1.4 Blood group

Welcome — one clear promise, one main button; “Skip” and “Sign in” for returning donors.

Sign in — only a mobile number; explains up front that hospitals see it only after you accept.

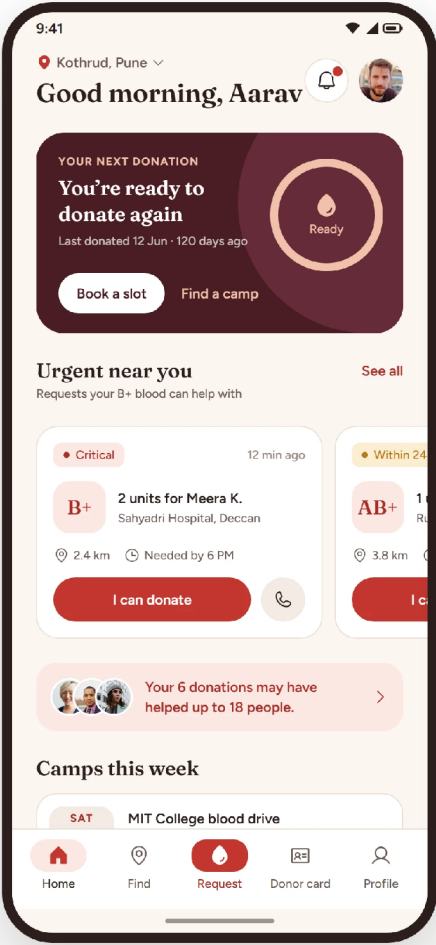
OTP — large keypad, resend timer, and “Change” if the number was wrong.

Blood group — tap one of 8 chips instead of typing; “I don’t know” is a valid answer.

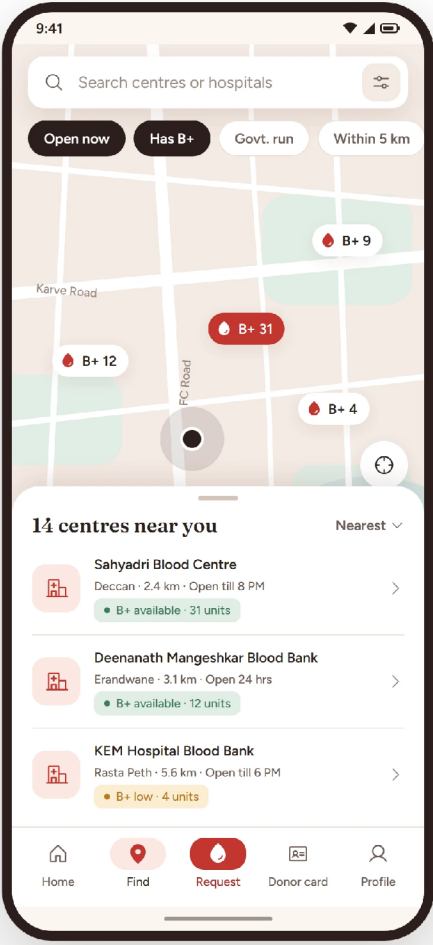
HCI PRINCIPLES APPLIED

- Visibility of status: Step 1 of 3
- User control
- Recognition over recall
- Error recovery

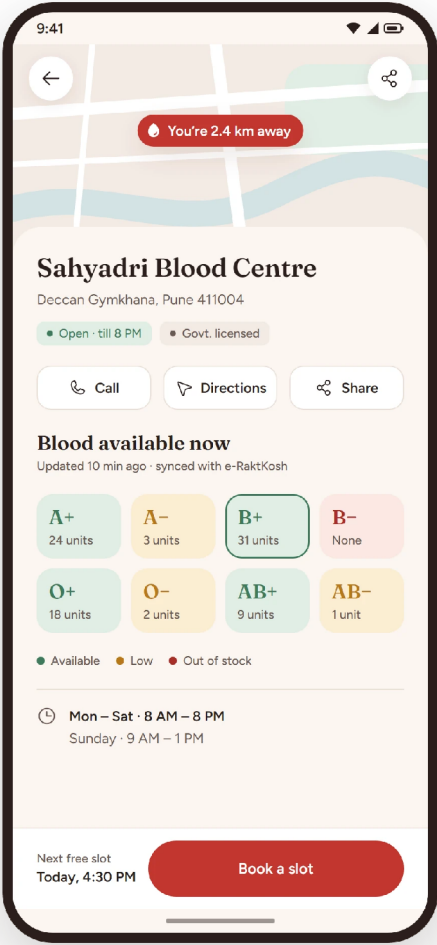
Know when you can donate, and where



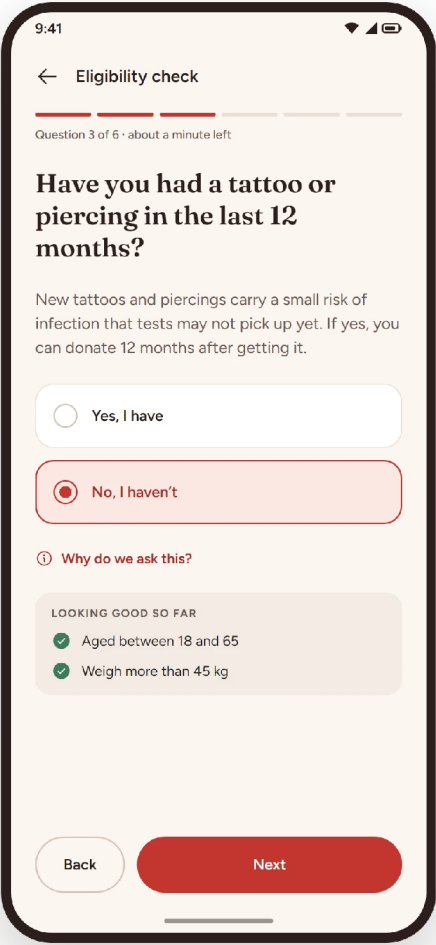
2.1 Home



2.2 Find centres



2.3 Centre details



2.4 Eligibility

Home — a status card says “You’re ready to donate again”; urgent requests show only patients the donor’s group can help.

Find — map plus list; filter chips such as “Has B+” and “Within 5 km”.

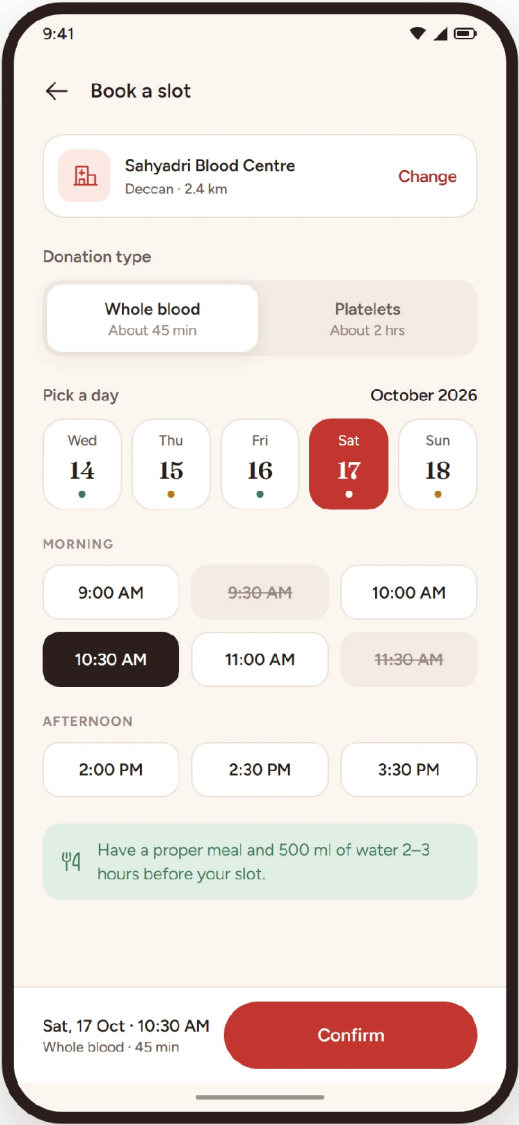
Centre — stock for all 8 groups with a text legend and “Updated 10 min ago”.

Eligibility — one question per screen, progress shown, and a “Why do we ask this?” note.

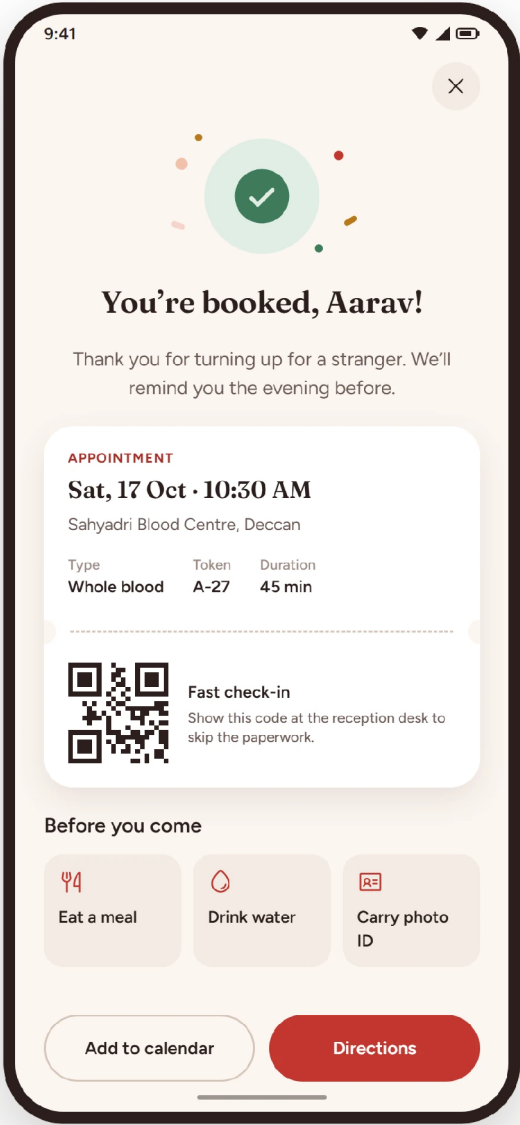
HCI PRINCIPLES APPLIED

- Match with the real world
- Hick’s law
- Help in context
- Error prevention

Booking a slot, then closure



3.1 Choose a slot



3.2 Booking confirmed

Donation type — a two-option segmented control shows how long each one takes (45 min vs 2 hrs).

Day and time — a date strip and slot grid grouped into Morning / Afternoon; full slots are struck through and cannot be picked.

Sticky summary — the chosen slot and the Confirm button stay at the bottom, within thumb reach.

Confirmation — token number, QR code for fast check-in, a “Before you come” checklist, Add to calendar and Directions.

HCI PRINCIPLES APPLIED

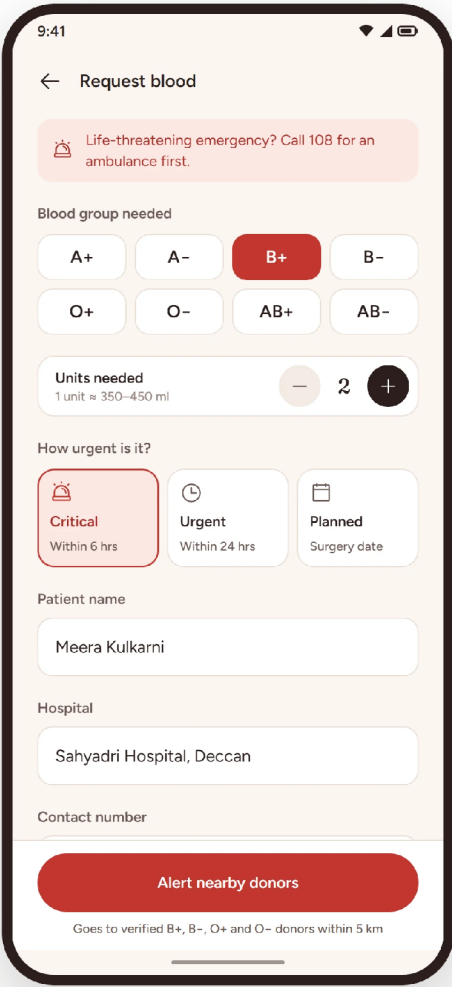
Gestalt proximity

Fitts’s law

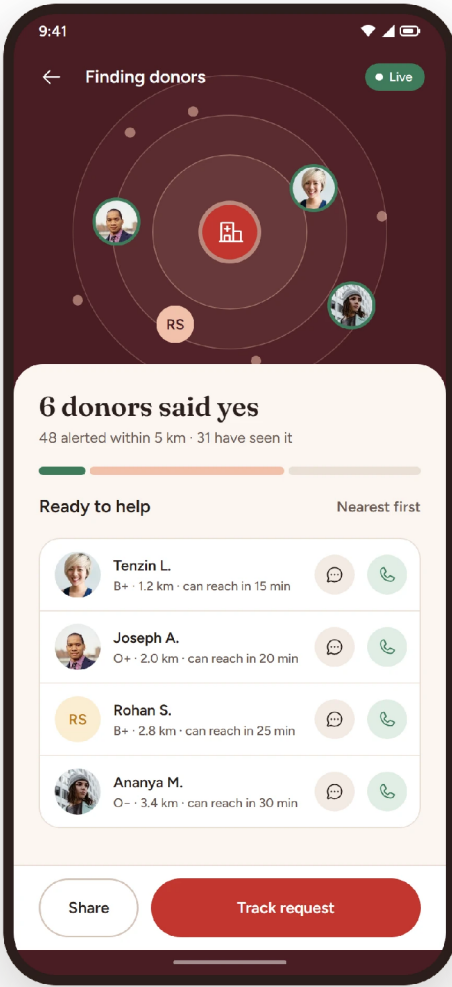
Closure (Shneiderman)

Reduce memory load

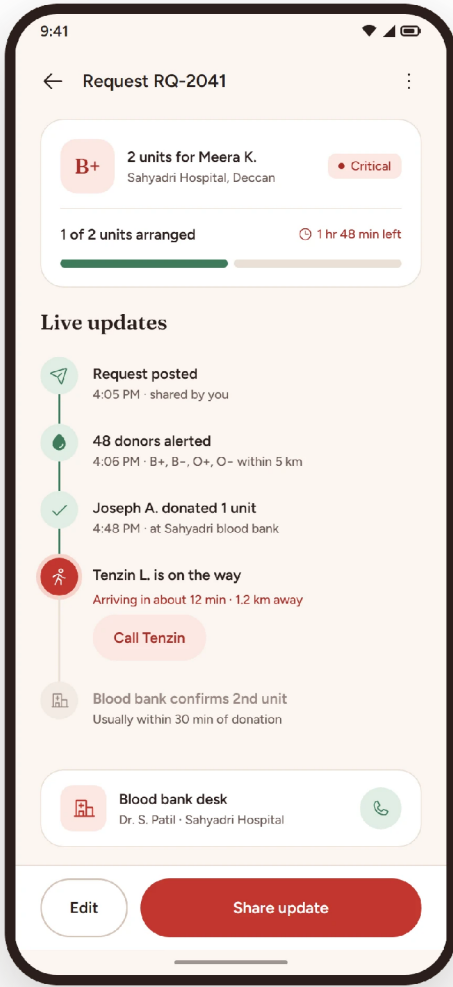
From request to donor arrival, in plain sight



4.1 Request blood



4.2 Matching donors



4.3 Request tracking

Safety first — a banner says to call 108 for an ambulance before anything else.

Compatibility done for you — the family picks the patient’s group; the app alerts every compatible group (B+, B-, O+, O-) and says so on the button.

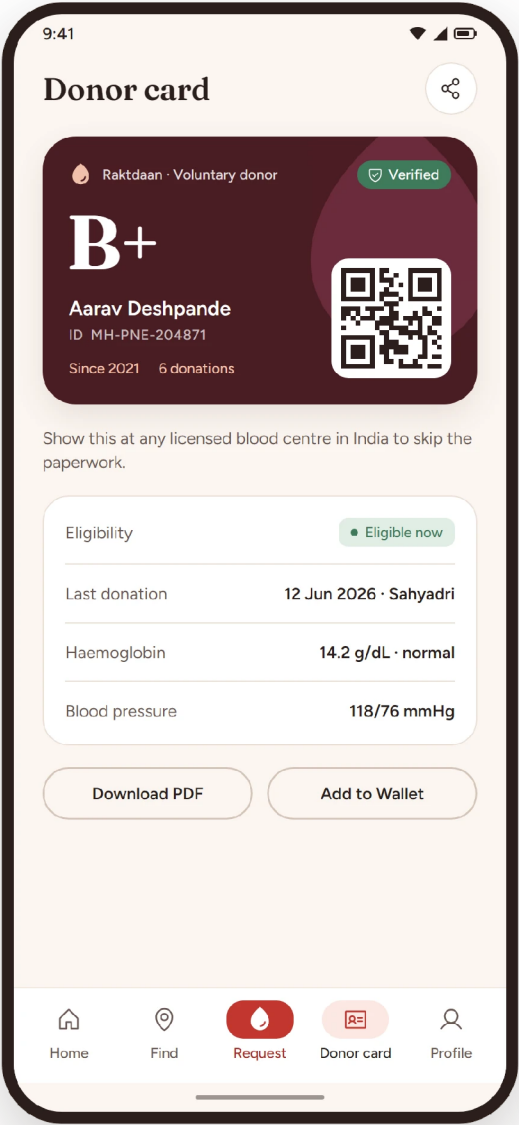
Live matching — a darker screen signals urgency: 48 alerted, 31 seen, 6 said yes, nearest first.

Tracking — “1 of 2 units arranged”, time left, and a timeline of every update.

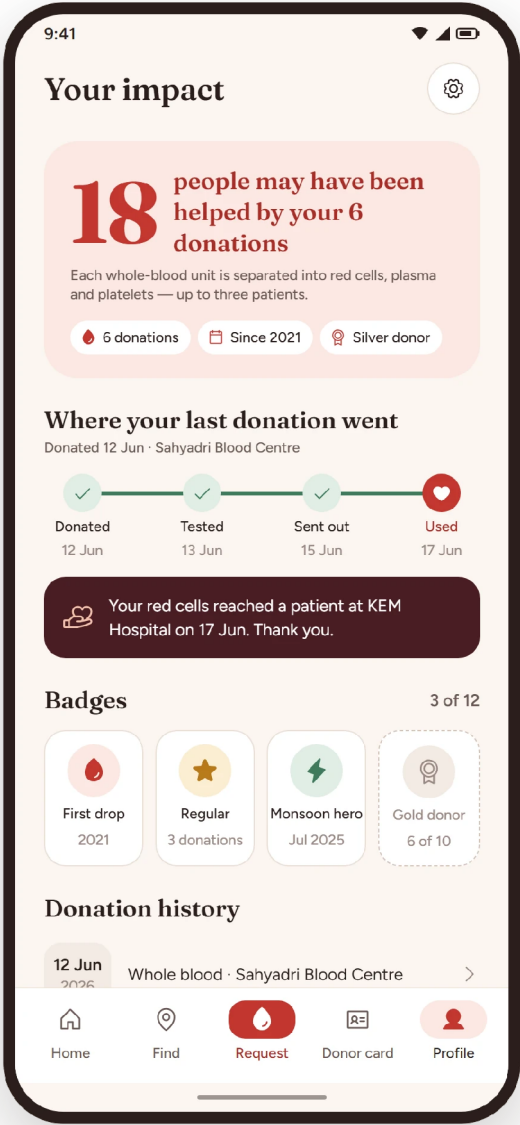
HCI PRINCIPLES APPLIED

- Visibility of system status
- Error prevention
- Emotional design
- Feedback

Reasons to come back



5.1 Donor card



5.2 Your impact

Digital donor card — verified badge, QR code, eligibility, last haemoglobin and blood pressure — no paperwork at the next centre.

Impact — “18 people may have been helped by your 6 donations”, with the reason: one unit becomes red cells, plasma and platelets.

Where your blood went — Donated → Tested → Sent out → Used, ending with the hospital that used it.

Badges and history — small rewards for regular donors, and a record of every donation.

HCI PRINCIPLES APPLIED

Feedback loop

Honest wording

Recognition

Motivation

Nielsen’s 10 heuristics, checked against the design

HEURISTIC	WHERE RAKTDAAN APPLIES IT
1 Visibility of system status	“Step 2 of 3”, “Question 3 of 6 · about a minute left”, live request timeline, stock “updated 10 min ago”
2 Match with the real world	+91 numbers, km, Pune hospitals, units in ml, NBTC donor rules, 108 ambulance
3 User control and freedom	Skip, Back on every step, “Change” number or centre, pause alerts any time
4 Consistency and standards	One component library; Android system bars and bottom navigation; main button always at the bottom
5 Error prevention	Eligibility check before booking; app works out compatible groups; full slots cannot be picked
6 Recognition rather than recall	Blood-group chips, date strip and slot grid, QR donor card instead of typing IDs
7 Flexibility and efficiency	“I can donate” right on Home; filter chips; Add to calendar; Request is one tap away
8 Aesthetic and minimalist design	One main action per screen; warm, calm palette; red only for actions and urgency
9 Help users recover from errors	Inline error text on inputs; “I don’t know my blood group”; OTP resend timer and “Change”
10 Help and documentation	“Why do we ask this?”, “Did you know?” tips, “Before you come” checklist

Accessibility check against WCAG 2.2

Colour contrast (WCAG AA: 4.5 : 1)

TEXT / BACKGROUND	USED FOR	RATIO	
Aa Primary text / canvas	Headings, body	15.0 : 1	Pass
Aa Secondary text / canvas	Supporting text	6.1 : 1	Pass
Aa White / brand red	Primary buttons	5.4 : 1	Pass
Aa Brand text / blush	Critical tag	5.7 : 1	Pass
Aa White / wine	Matching screen	14.1 : 1	Pass
Aa Primary text / white	Cards, inputs	16.1 : 1	Pass
Aa Brand text / canvas	Links: "Change", "Sign in"	6.3 : 1	Pass
Aa Secondary text / sunken	Neutral tag	5.5 : 1	Pass

Touch targets and semantics

- Main buttons are 56 px tall, icon buttons 48 × 48 px, bottom-bar items 76 × 58 px, blood-group chips 76 px tall.
- Status never relies on colour alone: every tag has a word, and the stock grid has a legend.
- Choices are real radio buttons and checkboxes (54 inputs); icon-only controls carry text labels (48 aria-labels), so screen readers can announce them.
- Visible focus ring on every control; animations switch off when "reduce motion" is on.

Result: main text, links and buttons meet WCAG AA contrast, and every touch target is well above the 24 px minimum (WCAG 2.5.8). Ratios are computed from the design tokens in raktdaan.html.

A clickable prototype with zero JavaScript

- **One file**, raktdaan.html: 1,430 lines of HTML and CSS, all 15 screens.
- **Navigation**: each screen is a <section id>; buttons are links; CSS **:target** shows the current screen.
- **State**: blood groups, slots, answers and filters are real inputs, styled with **:has(:checked)**.
- **Tokens**: CSS custom properties with the same names as the Figma variables; text styles as classes.
- **Icons**: 42 Phosphor icons in one inline SVG sprite (<symbol> + <use>).
- **Responsive**: on a laptop it shows a phone frame and a screen menu; on a real phone the page becomes the app.

```
/* 1 · navigation without JavaScript */
.screen { display: none; }
.screen:target { display: block; }

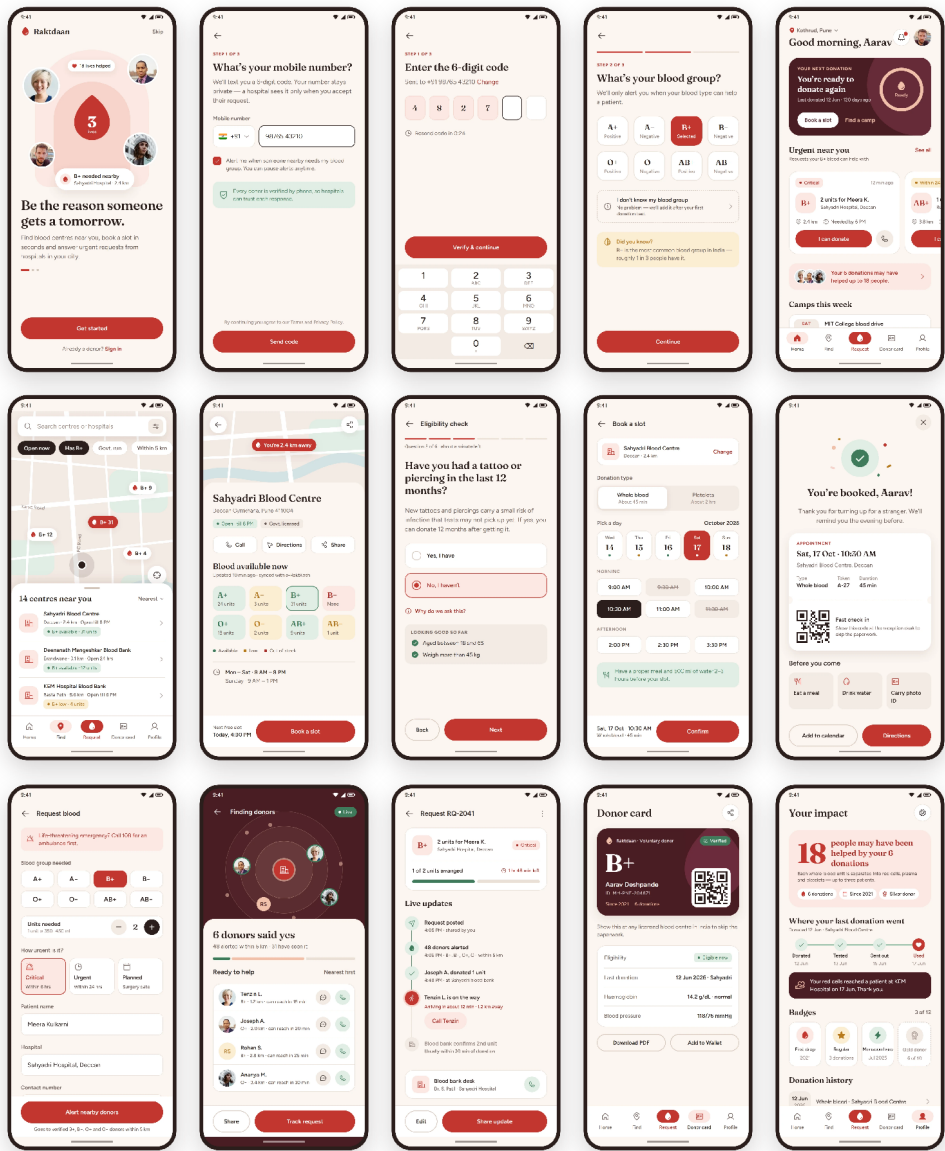
<a href="#slot" class="btn btn--white">Book a slot</a>

/* 2 · selected state from a real radio input */
.bg-chip:has(input:checked) {
  background: var(--brand);
  color: var(--text-on-brand);
}

<label class="bg-chip">
  <input class="sr" type="radio" name="my-group" value="B+">
  <b>B+</b> ...
</label>
```

From Figma to code without losing the design

IN FIGMA	IN HTML & CSS
Colour and radius variables	Custom properties: --brand , --bg-canvas , --radius-md
Text styles	Classes: .t-display , .t-headlinet-overline
Components and variants	Class + modifier: .btn--primary , .tag--urgent
Interactive states	:hover , :focus-visible , :has(:checked)
Prototype links	 + :target
Android frame, 412 × 917	.phone , 412 × 917 px; full screen on a real phone
Auto layout	Flexbox and grid helpers: .row , .col , .gap-16



All 15 screens, rendered from the HTML

CONCLUSION

What we built and what comes next

- Raktdaan covers both sides of blood donation — a donor's routine and a family's emergency — in **15 screens** and **5 flows**.
- One design system drives both the Figma file and the HTML & CSS build, so the design and the code match.
- Our heuristic and WCAG review confirmed clear system status, error prevention, recognition over recall, **AA contrast** for text and buttons, and **large touch targets**.

PPT

This deck: the problem, users, flows, design system and evaluation.

Figma design

Variables, text styles, components, icons and 15 linked screens.

HTML & CSS code

raktdaan.html — opens in any browser, no install, no JavaScript.

Future work

Usability tests with donors and families (task time, SUS)

Hindi and Marathi

Dark mode

Live stock from e-RaktKosh

Push alerts

Blood-centre staff dashboard

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Thank you

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12. MDN Web Docs, "":target" and "":has()" CSS pseudo-classes, Mozilla.

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